

GROCERY SHOPPING LIST MANAGEMENT

Using Singly Linked List in C++

1. Problem Statement

Managing grocery items manually can be difficult when items need to be added, displayed, searched, or deleted frequently. A simple menu-driven application is required to maintain grocery items efficiently using a singly linked list.

The system should allow the user to add multiple grocery items, display the complete shopping list, search for a particular item, and delete an item when it is no longer required.

2. Objectives

- Maintain grocery items using a singly linked list.
- Add multiple grocery items dynamically.
- Display all items in the list.
- Search for a particular grocery item.
- Delete an unwanted grocery item.
- Understand practical linked-list operations.

3. Main Operations

Add Items: Creates nodes and inserts items at the end.

Display List: Traverses and displays all items.

Search Item: Searches for a required item.

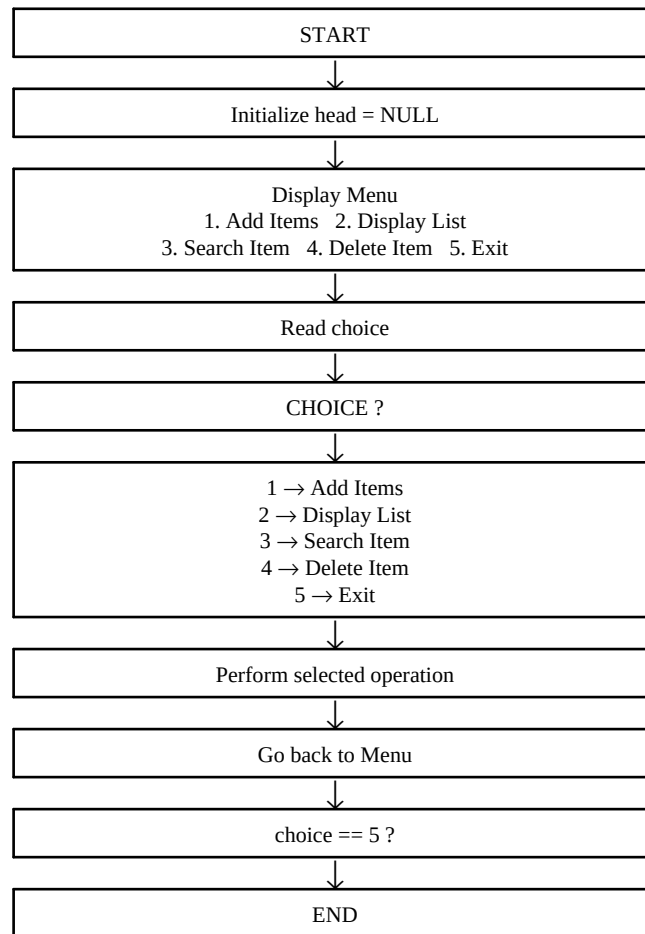
Delete Item: Removes the selected item and adjusts links.

Exit: Terminates the program.

ALGORITHM

1. Start the program.
2. Initialize head = NULL.
3. Display the menu: Add Items, Display List, Search Item, Delete Item, Exit.
4. Read the user's choice.
5. For choice 1, read the number of items and insert each item at the end of the linked list.
6. For choice 2, check whether the list is empty. If not, traverse and display all items.
7. For choice 3, read the item, traverse the list, and compare each node. Display found or not found.
8. For choice 4, read the item, search for its node, adjust links, delete the node, and display the result.
9. For choice 5, display the exit message and terminate.
10. For any other choice, display an invalid-choice message.
11. Repeat the menu until choice 5 is selected.
12. Stop the program.

FLOWCHART



Operation details: Add = create and insert nodes; Display = traverse list; Search = compare items; Delete = adjust links and delete node.

Problems Output Debug Console Terminal Ports

```
PS D:\siddanth> cd 'd:\siddanth\output'
PS D:\siddanth\output> & .\'liked list 3.exe'
```

===== Grocery Shopping List =====

1. Add Items
2. Display List
3. Search Item
4. Delete Item
5. Exit

Enter your choice: 1

How many grocery items do you want to add? 4

Enter grocery item 1: sugar

Enter grocery item 2: corn

Enter grocery item 3: rice

Enter grocery item 4: wheat

Items added successfully.

===== Grocery Shopping List =====

1. Add Items
2. Display List
3. Search Item
4. Delete Item
5. Exit

Enter your choice: 2

Grocery Shopping List:

- sugar
- corn
- rice
- wheat

===== Grocery Shopping List =====

1. Add Items
2. Display List
3. Search Item
4. Delete Item
5. Exit

Enter your choice: 3

Enter item to search: sugar

===== Grocery Shopping List =====

1. Add Items
2. Display List
3. Search Item
4. Delete Item
5. Exit

Enter your choice: 3

Enter item to search: sugar

Item found in the shopping list.

===== Grocery Shopping List =====

1. Add Items
2. Display List
3. Search Item
4. Delete Item
5. Exit

Enter your choice: 4

Enter item to delete: wheat

Item deleted successfully.

===== Grocery Shopping List =====

1. Add Items
2. Display List
3. Search Item
4. Delete Item
5. Exit

Enter your choice: 2

Grocery Shopping List:

- sugar
- corn
- rice

===== Grocery Shopping List =====

1. Add Items
2. Display List
3. Search Item
4. Delete Item
5. Exit